Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/14/2008
PI Name:	Ullrich, Robert Ph.D.		
Project Title:	NSCOR: Radiation Leukemogenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bullrich@utmb.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	409-747-1935
Organization Name:	University of Texas Medical Branch		
PI Address 1:	301 University Blvd		
PI Address 2:	Comprehensive Cancer Center, MS 1048		
PI Web Page:			
City:	Galveston	**State:**	TX
Zip Code:	77555-5302	**Congressional District:**	14
Comments:	NOTE: PI moved to UTMB from Colorado State University in late 2008 (6/2009)		
Project Type:	Ground	**Solicitation / Funding Source:**	NSCOR 03-OBPR-02
Start Date:	10/01/2003	**End Date:**	07/30/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/30/2009 per J. Dardano/JSC (10/2009) NOTE: Received NCE to 12/31/2008 (from 9/30/2008) per J. Dardano/JSC (9/2008)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bailey, Susan (Colorado State University) Bedford, Joel (Colorado State University) Belmont, John (Baylor College of Medicine) Cai, Wei-Wen (Baylor College of Medicine) Ehrhart, E.J. (Colorado State University) Nagarajan, Lalitha (U.T., M.D. Anderson. Cancer Center) Story, Michael (University of Texas, M. D. Anderson Cancer Center) Weil, Michael (U.T., M.D. Anderson. Cancer Center)		
Grant/Contract No.:	NAG9-1569		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The goal of this program is to provide information required to develop a rational scientific basis for estimation of risks for leukemogenesis in humans from exposure to radiation during space flight. This will be accomplished using several coordinated research activities. A key activity will be to develop quantitative data on the relative leukemogenic effectiveness of specific HZE particles and of high-energy protons compared with gamma rays in a mouse model of radiation-induced acute myeloid leukemia (AML). Extrapolation of risks from this mouse model to risks in humans requires additional information, and a major component of research activities in this program are designed to provide such information. A major research effort of this program is focused on studies of cellular and molecular mechanisms and processes involved in radiation-induced AML. These studies will not only improve the basic understanding of the pathogenesis of AML but will also provide information on cellular and molecular biomarkers that may be useful in predicting individuals at increased risk. These activities will focus on the use of molecular and molecular cytogenetic approaches, the development of new model systems, and comparative analyses of murine and human AML to facilitate the translation of results obtained in mice to human AMLs. Problems of extrapolation are not limited to biological factors. There are clear dosimetric differences that also must be considered when extrapolating data obtained from mouse studies to human risks. To address these issues this project also has an important physics and dosimetry component that includes modeling of dose to target cells in murine and human bone marrow.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research provides information on risks from exposure to ionizing radiation and mechanisms of carcinogenesis. It has also provided insight into potential approaches to chemoprevention of radiation damage.
Task Progress:	Key Findings: a) In mice, 1 GeV/n 56Fe ions do not appear to be substantially more effective than gamma rays for induction of AML b) In contrast, 56Fe irradiated mice had much higher risk for hepatocellular carcinoma than gamma ray irradiated mice c) Chromosome 2 deletions detected in bone marrow cells one month post-irradiation may be a biomarker for AML risk d) There are no apparent differences between 56Fe and gamma ray induced AML at the cytogenetic or molecular level
Bibliography Type:	Description: (Last Updated: 06/10/2025)
Articles in Peer-reviewed Journals	Williams ES, Stap J, Essers J, Ponnaiya B, Luijsterburg MS, Krawczyk PM, Ullrich RL, Aten JA, Bailey SM. "DNA double-strand breaks are not sufficient to initiate recruitment of TRF2." Nat Genet. 2007 Jun;39(6):696-8; author reply 698-9. PMID: 17534357 , Jun-2007
Articles in Peer-reviewed Journals	Carsten RE, Bachand AM, Bailey SM, Ullrich RL. "Resveratrol reduces radiation-induced chromosome aberration frequencies in mouse bone marrow cells." Radiat Res. 2008 Jun;169(6):633-8. PMID: 18494544 , Jun-2008
Articles in Peer-reviewed Journals	Ding LH, Xie Y, Park S, Xiao G, Story MD. "Enhanced identification and biological validation of differential gene expression via Illumina whole-genome expression arrays through the use of the model-based background correction methodology." Nucleic Acids Res. 2008 Jun;36(10):e58. Epub 2008 May 1. PMID: 18450815 , Jun-2008
Articles in Peer-reviewed Journals	Hagelstrom RT, Askin KF, Williams AJ, Ramaiah L, Desaintes C, Goodwin EH, Ullrich RL, Bailey SM. "DNA-PKcs and ATM influence generation of ionizing radiation-induced bystander signals." Oncogene. 2008 Aug 4. [Epub ahead of print] PMID: 18679419 , Aug-2008
Articles in Peer-reviewed Journals	Peng Y, Brown N, Finnon R, Warner CL, Liu X, Genik PC, Callan MA, Ray FA, Borak TB, Badie C, Bouffler SD, Ullrich RL, Bedford JS, Weil MM. "Radiation Leukemogenesis in Mice: Loss of PU.1 on Chromosome 2 in CBA/Ca and C57BL/6 Mice after Irradiation with 1 GeV/n 56Fe Ions, X- or gamma-rays. Part I. Experimental Observations." Submitted Radiation Research, August 2008. , Aug-2008
Articles in Peer-reviewed Journals	Peng Y, Borak TB, Bouffler SD, Ullrich RL, Weil MM, Bedford JS. "Radiation Leukemogenesis in Mice: Loss of PU.1 on Chromosome 2 in CBA/Ca and C57BL/6 Mice after Irradiation with 1 GeV/n 56Fe Ions or gamma-Rays. Part II. Theoretical Considerations Based on Microdosimetry and the Initial Induction of Chromosome Aberrations." Submitted Radiation Research, August 2008. , Aug-2008
Articles in Peer-reviewed Journals	Weil MM, Bedford JS, Erhart EJ, Ray FA, Genik PC, Fallgren C, Ullrich RL. "Incidence of acute myeloid leukemia and hepatocellular carcinoma in 1 GeV/n 56Fe irradiated mice." Submitted Radiation Research, August 2008. , Aug-2008
Patents	61/036,399. Pending, March 2008. Mar-2008 Carsten R, Ullrich R. "Reserveratrol pre and post exposure radioprotectant."
Patents	61/050,251. Pending, May 2008. May-2008 Carsten R, Ullrich R. "Grape as a Radioprotectant."